

Synergizing Knowledge Graphs, Voice HCI, and Decentralized Webs for Intelligent Bookstores

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Project Proposal Report
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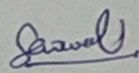
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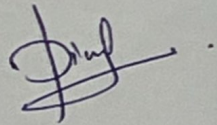
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lankan readers, students, and emerging authors currently experience a fragmented digital literary environment where reviews, comments, and reading notes are usually locked inside individual online bookstore platforms. These limits open literary discussion, user ownership of content, and cross-platform academic collaboration.

This research proposes **FedBook-Sem**, the federated social component of an intelligent online bookstore platform. The component integrates social media–style interactions such as reviews, comments, likes, and shares into an online bookstore using **ActivityPub** and the **W3C Web Annotation Data Model**. It also introduces semantic annotation support for collaborative reading and uses **Neo4j** to model social relationships such as follows, replies, likes, boosts, and annotation threads.

By transforming an online bookstore into a participant in the decentralized social web, the proposed system supports user-owned literary discourse, interoperable academic discussion, and open knowledge exchange.

Keywords: ActivityPub, Federated Social Graph, Web Annotation, Neo4j, Federated Literary Discourse.

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LIST OF ABBREVIATIONS

- **API** – Application Programming Interface
- **JSON-LD** – JavaScript Object Notation for Linked Data
- **NLP** – Natural Language Processing
- **RDF** – Resource Description Framework
- **SDG** – Sustainable Development Goals
- **URI** – Uniform Resource Identifier
- **W3C** – World Wide Web Consortium

1. INTRODUCTION

1.1 Background and Context

Sri Lanka’s digital literary ecosystem currently offers readers, students, and emerging authors an experience that is fragmented, platform-bound, and culturally limited. Traditional online bookstores focus mainly on catalogue browsing, transactions, ratings, and isolated reviews. Although these features support book sales, they do not support meaningful open literary discourse. Reviews, comments, reading notes, and annotations remain locked inside the bookstore platform, preventing them from being shared, reused, or discussed across wider academic and reader communities.

Current online literary platforms often represent “walled gardens” where user reviews and annotations remain captive within a single site. These limits open academic discourse and undermine data sovereignty. The broader research project intends to address multiple limitations in Sri Lankan online bookstores through four integrated technological components: a Stylistic-Semantic Knowledge Graph, a phonetic and voice-driven search framework, a federated social layer, and an adaptive pricing ecosystem.

This proposal focuses on **FedBook-Sem**. The sub-objective of FedBook-Sem is to design and implement a **Federated Social Graph architecture** that integrates social-media-style interactions such as reviews, comments, likes, and shares into an online bookstore using **ActivityPub** and the **W3C Web Annotation Data Model**, thereby enabling decentralized, user-owned literary discourse.

This problem is especially significant in the Sri Lankan context because literary discussion often extends beyond product feedback. It includes educational interpretation, cultural reading, classroom learning, academic analysis, and community discussion. When these interactions are locked into one bookstore platform, the social value of literature is reduced to a private commercial feature instead of a reusable public knowledge resource.

To solve this, FedBook-Sem reimagines the bookstore as part of the **decentralized social web**. Instead of keeping reviews and discussion inside one database table, the system treats them as portable, interoperable, standards-based social objects. In this way, the bookstore becomes not just a seller of books, but a participant in a broader ecosystem of literary exchange, collaborative reading, and open discourse.

8. How useful would it be to highlight specific sentences in an online book and discuss them directly with your friends across different social platforms? | [Copy chart](#)

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87 responses

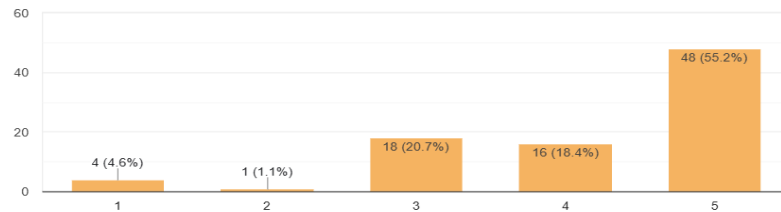


Figure 1: Usefulness of Cross-Platform Sentence Discussions

Usefulness of Cross-Platform Sentence Discussions: Furthermore, when respondents were asked how useful it would be to highlight specific sentences in an online book and discuss them directly with friends across different social platforms, the majority showed a strong positive response. Specifically, **55.2%** selected **5**, indicating the highest level of usefulness, while **18.4%** selected **4**. In addition, **20.7%** selected **3**, showing moderate interest. Only a small percentage selected **1** or **2**. These findings suggest that most users find this feature highly useful, supporting the need for collaborative and socially connected reading experiences.

9. Would you like it if book reviews from social media were also integrated into the website, allowing you to choose books based on those responses? | [Copy chart](#)

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87 responses

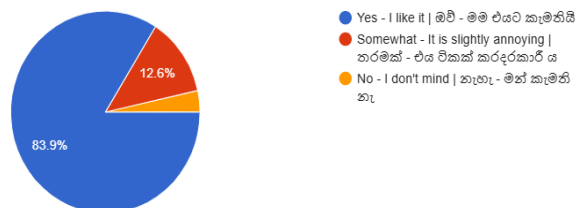


Figure 2: User Survey Findings on Social Interaction Features

Interest in Integrating Social Media Reviews: Furthermore, when respondents were asked whether they would like book reviews from social media to be integrated into the website to help them choose books, the overwhelming majority responded positively. Specifically, **83.9%** stated **“Yes – I like it”**, while **12.6%** selected **“Somewhat”**. Only a very small percentage chose **“No – I don’t mind.”** These findings indicate that most users are highly interested in having social media reviews integrated into the platform, as it can support better decision-making when selecting books.

1.2 Literature Review

Most traditional online bookstores and digital commerce systems are designed around centralized data ownership. Reviews and comments are typically stored only inside the platform that hosts them. Their structures are usually optimized for basic product feedback rather than open communication, interoperability, or academic reuse. Existing systems are dominated by collaborative filtering, keyword matching, strict text search, and centralized engagement models. They do not support federated literary-social interactions.

In contrast, decentralized social web technologies propose a different model. **W3C ActivityPub** is an open standard that supports federated social networking through inbox/outbox communication among distributed actors. It allows one server to publish social activities and another server to receive and process them. This model is widely associated with federated platforms such as Mastodon. FedBook-Sem will implement an **ActivityPub-Based Social Interaction Layer** where both users and books function as first-class actors, and where book reviews are published as signed JSON-LD activities that can be followed, liked, replied to, and shared across federated platforms.

A second key area is the **W3C Web Annotation Data Model**, which provides a structured way to represent annotations on digital content. This is particularly useful in the context of books, where readers may want to highlight text, add notes to a paragraph, discuss a sentence, or create collaborative reading threads. FedBook-Sem will integrate a **Semantic Annotation System for Collaborative Reading** to support sentence-level and paragraph-level annotations that are portable, exportable, and interoperable with external scholarly tools.

Another relevant body of work concerns graph databases and social network modelling. Social relationships such as follows, likes, replies, boosts, and discussion threads are naturally graph-structured rather than table-structured. Relational databases can store them, but deeper traversals become increasingly difficult and inefficient. Therefore, **Neo4j** is proposed to model these complex relationships, and graph-based traversal enables queries such as “books liked by people I follow across federated servers,” which would be impractical in relational SQL databases.

The literature therefore shows that federation, annotation, and graph-based social modeling are powerful concepts. However, what remains largely missing is their combined application within an **online bookstore** in a way that supports **literary discourse**, **academic collaboration**, and **user-owned social reading**. That integration is the core contribution of FedBook-Sem.

Table 1: Summary of Existing Literature and System Limitations

Reference	Research Focus / Proposed System	Key Methodology	Identified Limitation (Addressed by FedBook-Sem)
Traditional Online Bookstores	Site-bound reviews and rating systems	Centralized databases and product review tables	Reviews and comments are locked inside a single platform
Federated Social Platforms	Distributed social networking	ActivityPub inbox/outbox communication	Not specifically adapted for book-centred literary discourse
Annotation Standards	Portable scholarly annotations	W3C Web Annotation Data Model	Often not integrated into bookstore ecosystems
Graph Databases	Social relationship modeling	Social relationship modeling	Rarely applied to bookstore reviews, annotations, and reading communities
Existing Sri Lankan Book Platforms	Transaction-focused digital book sales	Conventional web application logic	Lack interoperable literary-social interaction and annotation portability

1.3 Research Gap

The main research gap lies in the absence of a practical, bookstore-integrated system that combines **federated social networking, portable literary annotation, and graph-based modeling of literary interaction**.

A major limitation in existing systems is **centralized discourse**, where reviews and reading notes remain trapped within the platform instead of being shared through open web standards. This means that readers cannot own their discourse, institutions cannot reuse discussion easily, and literary conversations remain fragmented.

There are several specific gaps. First, current online bookstores do not support **cross-platform literary interaction**. A review written inside one bookstore stays there and cannot naturally participate in wider open-web discussion. Second, traditional systems do not treat reviews, comments, likes, and shares as **interoperable social objects**. Instead, they treat them as local site content. Third, sentence-level and paragraph-level annotations are rarely modelled as portable academic objects inside bookstores. Fourth, relational databases are poorly suited for deep recursive social-literary queries, limiting discovery patterns such as following chains of influence, annotation networks, and community reading behaviour.

FedBook-Sem addresses this gap by combining ActivityPub, Web Annotation, and Neo4j into a unified bookstore social layer. The novelty of this component lies in eliminating the traditional “walled garden” approach and reimagining the online bookstore as an active participant in the decentralized social web, promoting user data sovereignty, open knowledge exchange, and interoperable academic collaboration.

Table 2: Comparison of the existing research methods and proposed method

Technologies and Methods	[1] ActivityPub Spec (2018)	[2] Mastodon Federation	[3] Web Annotation Model	[4] Neo4j Graph Research	[5] Decentralized Social Network Studies	Proposed Method
ActivityPub based federation	✓	✓	✗	✗	✓	✓
Decentralized social interaction	✓	✓	✗	✗	✓	✓
Book review as federated object	✗	✗	✗	✗	✗	✓
Cross-platform social interaction	✓	✓	✗	✗	✓	✓
Semantic text annotation	✗	✗	✓	✗	✗	✓
Sentence/paragraph level annotation	✗	✗	✓	✗	✗	✓
Graph database for social relationships	✗	✗	✗	✓	✓	✓
Social graph traversal queries	✗	✗	✗	✓	✗	✓
Integration with online bookstore	✗	✗	✗	✗	✗	✓
Federated literary discussion platform	✗	✗	✗	✗	✗	✓

2. OBJECTIVE

2.1 Main Objective

To design and implement a **Federated Social Graph architecture** that integrates decentralized social interactions into an online bookstore using open web standards such as ActivityPub and the W3C Web Annotation Data Model.

2.2 Specific Objectives

The specific focus of this individual component, **FedBook-Sem**, is to design and implement a **Federated Social Graph architecture** that integrates social media–style interactions such as reviews, comments, likes, and shares into an online bookstore using open web standards such as ActivityPub and the W3C Web Annotation Data Model, enabling decentralized, user-owned literary discourse.

To achieve this component successfully, the research will accomplish the following specific objectives:

1. **To implement an ActivityPub-Based Social Interaction Layer**
This layer will implement the W3C ActivityPub inbox/outbox architecture where both users and books function as first-class actors. Reviews will be published as signed JSON-LD activities and behave like social media posts that can be followed, liked, replied to, and shared across federated platforms while synchronizing all interactions back to the bookstore system.
2. **To develop a Federated Reviews, Comments, and Likes Model**
This objective replaces traditional site-locked comment sections with a federated discourse model where reviews, comments, likes, and shares are treated as interoperable social objects on the open web. This enables cross-platform engagement while preserving data ownership and reducing platform lock-in.
3. **To integrate a Semantic Annotation System for Collaborative Reading**
This objective will incorporate the W3C Web Annotation Data Model to support sentence-level and paragraph-level annotations within books. These annotations will be portable, exportable, and interoperable with external scholarly tools, enabling textbook study, academic discussion, and collaborative learning beyond a single platform.
4. **To deploy a Graph Database Implementation using Neo4j**
This objective will use Neo4j to model complex social relationships such as follows, replies, likes, boosts, and annotation threads. Graph-based traversal will enable efficient querying of recursive literary-social structures that are impractical to represent and analyse efficiently using relational SQL databases.

3. METHODOLOGY

3.1 Research Approach

This research adopts a **System Development and Evaluation Methodology** because the main goal is to design and implement a novel federated social architecture to solve a real-world problem in digital literary discourse. The project does not merely analyse an existing issue; it engineers a new technical solution and evaluates how effectively it addresses platform lock-in, interoperability, and collaborative reading.

This methodology is combined with an **experimental and empirical validation approach**. A baseline of traditional closed-bookstore interaction will be used for comparison, and the FedBook-Sem component will be tested for federation success, data portability, graph efficiency, and usability.

The project will be carried out through the following phases:

1. Domain and Standards Analysis

This initial phase focuses on understanding decentralized web protocols, literary-social interaction patterns, academic annotation requirements, and graph data modeling. ActivityPub specifications, Web Annotation structures, and bookstore review behaviours will be analyzed.

2. Social Architecture Design

In this phase, the actor model for users and books will be designed. Inbox and outbox federation logic, JSON-LD activity structures, semantic annotation models, and the Neo4j schema will be planned carefully.

3. Component Implementation

The ActivityPub-based social interaction layer, federated review/comment model, semantic annotation module, and graph database layer will be implemented and integrated into the broader intelligent bookstore platform.

4. Federation and Interoperability Testing

The implemented component will be tested using federated scenarios such as follow requests, review publication, likes, replies, boosts, and annotation exchange. External and simulated endpoints will be used to evaluate standards compliance and communication reliability.

5. Graph Evaluation and Optimization

Recursive queries on the Neo4j social graph will be tested for performance and correctness. Query pathways such as community influence, annotation discussion chains, and cross-server literary-social traversal will be optimized.

6. Final Documentation and Reporting

Finally, the research findings, evaluated results, system limitations, and future improvements will be documented comprehensively in the final dissertation and research outputs.

3.2 Technical Methods and Algorithms

The FedBook-Sem system uses a combination of standards-based federation, semantic representation, and graph-based social modeling.

- **ActivityPub Federation Logic**
The system will implement **Inbox/Outbox architecture** using ActivityPub. Users and books are treated as actors. Activities such as Create, Follow, Like, Reply, and Announce will be handled through standards-compliant federation flows.
- **Signed JSON-LD Review Publishing**
Book reviews will be represented as signed JSON-LD activities rather than plain local database comments. This allows them to behave like social objects that can travel across federated systems while preserving structure and provenance.
- **Federated Interaction Synchronization**
Likes, comments, replies, and shares from external federated systems will be synchronized back into the bookstore environment so that the local interface remains aware of broader discourse.
- **Semantic Annotation Encoding**
Sentence-level and paragraph-level annotations will be encoded according to the **W3C Web Annotation Data Model**. This enables collaborative reading and scholarly portability.
- **Graph-Based Social Traversal**
Neo4j will model relationships such as follows, likes, replies to, boosts, annotates, and mentions. Graph traversal queries will support recursive discovery patterns and literary community analysis.

3.3 Tools and Platforms with Justification

To implement FedBook-Sem effectively, the following tools and platforms are selected:

- **Python or Node.js Backend Frameworks**
Suitable for implementing APIs, federation endpoints, serialization logic, background jobs, and asynchronous communication.
- **ActivityPub / JSON-LD Libraries**
Required for generating, validating, signing, and transmitting federated activity objects.
- **Neo4j Graph Database**
Chosen because Neo4j is essential for modeling follows, replies, likes, boosts, and annotation threads, and for enabling efficient recursive querying of literary-social structures.
- **MongoDB or PostgreSQL (supporting storage)**
Useful for storing metadata, session information, platform configuration, and auxiliary content where graph storage is not necessary.
- **W3C Web Annotation Compatible Schemas**
Necessary to create structured, exportable annotations that remain interoperable.
- **Docker / Cloud Deployment Environment**
Useful for simulating multiple federated nodes, deploying inbox/outbox endpoints, and testing cross-instance communication.

3.4 Data Collection and Ethical Considerations

For successful development and validation of the FedBook-Sem framework, several categories of data are required.

Firstly, **sample literary-social interaction data** will be prepared. This includes example reviews, comments, likes, replies, boosts, and user-follow relationships that simulate real online bookstore behaviour.

Secondly, **annotation scenarios** will be created for collaborative reading. These scenarios will include sentence-level annotations, paragraph-level notes, interpretive comments, classroom discussion examples, and reading reflections linked to specific passages.

Thirdly, **graph relationship datasets** will be modelled. These will represent realistic user behaviours such as following a reviewer, replying to a review, liking an annotation, sharing a discussion thread, or discovering books through a social circle.

Ethical considerations are highly important because the component works with user-generated content and distributed communication. If human participants are involved in testing, **informed consent** will be obtained before collecting any interaction data. Any personally identifiable information will be anonymized wherever required.

In addition, federated systems introduce unique ethical considerations regarding **data portability**, **content visibility**, **cross-server copying**, and **user control**. Therefore, the research design will explicitly consider how literary content is shared, who can view it, and how the system can respect deletion or privacy expectations as much as technically feasible within a federated model.

3.5 Validation Strategy and Performance Metrics

The validation strategy for FedBook-Sem focuses on four major areas:

1. **Federation Reliability**

Measures whether reviews, likes, replies, follows, and shares are correctly transmitted and received across federated endpoints.

2. **Annotation Portability and Integrity**

Measures whether sentence-level and paragraph-level annotations preserve structure and meaning when exported, shared, or synchronized.

3. **Graph Query Performance**

Measures how efficiently Neo4j can traverse recursive literary-social relationships.

4. **User Task Success**

Measures whether users can successfully create reviews, reply, like, annotate text, and discover literary discussions.

3.6 Evaluation Plan

To ensure the proposed FedBook-Sem component meets its research objectives, it will be evaluated using the following metrics and methods.

Criteria	Metric	Method	Target
Federation Reliability	Successful Activity Delivery Rate	Test follow, create, like, reply, and announce interactions across federated nodes	$\geq 90\%$
Annotation Interoperability	Annotation Reconstruction Accuracy	Export and re-import annotations using W3C-compatible formats	$\geq 90\%$
Graph Performance	Average Query Response Time	Test recursive Neo4j queries under simulated load	< 2 seconds
Usability	Task Completion Rate	User testing for review posting, liking, replying, and annotating	$S \geq 85\%$ US Score $\geq 80/100$
Standards Compliance	Protocol Conformance Level	Validate JSON-LD activity structures and ActivityPub flow correctness	High compliance

Table 3: Evaluation Metrics & Methods

3.7 Expected Results & Real-World Use

Upon successful completion, the FedBook-Sem component is expected to:

- Replace platform-locked review systems with interoperable federated discourse.
- Allow book reviews to behave like social media posts that can be followed, liked, replied to, and shared across federated platforms.
- Support collaborative reading through portable semantic annotations.
- Model literary-social interaction efficiently using Neo4j graph structures.
- Promote user data sovereignty and open academic knowledge exchange.

In real-world use, this component can be adopted by online bookstores, academic publishers, educational reading systems, and digital humanities platforms. It can also create new kinds of literary engagement, such as public reading communities, annotation-driven classroom learning, and federated academic discussion around books.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The overall goal of the broader research is to create an **Intelligent Federated Bookstore Platform** that improves literary discovery, social interaction, accessibility, and dynamic market behavior in Sri Lanka. The proposed platform brings together four different technologies: **SSKG-SL** for stylistic-semantic recommendation, **PhonoLex-SL** for phonetic and voice-based search, **FedBook-Sem** for federated reviews and annotation, and **MusePrice** for adaptive pricing.

Within this ecosystem, FedBook-Sem is responsible for handling decentralized literary discourse and collaborative reading.

retrieval.

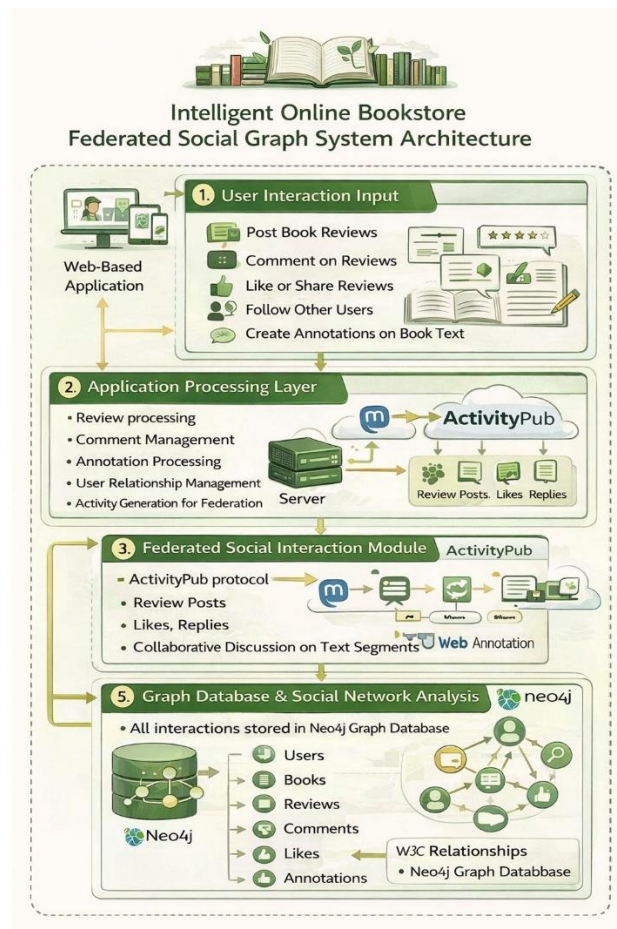


Figure 3: System Architecture

4.2 Detailed Component Breakdown (FedBook-Sem)

To implement decentralized literary-social functionality effectively, the FedBook-Sem component is divided into several interconnected modules:

User and Book Actor Layer

Defines both users and books as first-class actors in the ActivityPub model.

ActivityPub Inbox Module

Receives federated activities such as follows, likes, replies, and shared reviews from external platforms or other nodes.

ActivityPub Outbox Module

Publishes activities generated within the bookstore to federated recipients and external platforms.

Federated Review and Comment Engine

Converts local reviews and comments into signed JSON-LD activity objects and synchronizes responses back to the bookstore.

Semantic Annotation Module

Creates and stores sentence-level and paragraph-level annotations following the W3C Web Annotation Data Model.

Neo4j Social Graph Layer

Stores follows, likes, boosts, replies, annotation threads, and user-book interaction networks.

Integration and Presentation Layer

Displays synchronized discussions, annotations, and social interactions inside the bookstore interface.

4.3 Component Interactions and Data Flow

The FedBook-Sem component follows a real-time social interaction flow:

1. A user writes a review, likes a post, replies to a review, shares a discussion, or creates an annotation.
2. The bookstore system represents the interaction as an ActivityPub-compatible object.
3. The object is signed and sent through the outbox to relevant federated recipients.
4. External responses and remote activities are received through the inbox.
5. The local platform updates the Neo4j graph with relationships such as follows, likes, replies, or annotation references.
6. Semantic annotations are linked to the relevant book sections and discussion threads.
7. The bookstore interface displays both local and federated literary-social activity in a unified view

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- **Primary End Users**

Sri Lankan readers, students, educators, researchers, and literary enthusiasts who wish to discuss, annotate, and share book-related ideas in a more open and collaborative digital environment.

- **Secondary Stakeholders**

Online bookstore owners, publishers, academic institutions, book clubs, and digital reading communities that benefit from interoperable literary discussion and enhanced user engagement.

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5.2 Functional Requirements

The system shall be able to perform the following functions:

1. The system shall represent both users and books as first-class actors.
2. The system shall publish reviews as ActivityPub-compatible JSON-LD social activities.
3. The system shall support follows, likes, replies, and shares for literary interactions.
4. The system shall synchronize literary-social interactions with federated platforms.
5. The system shall support sentence-level and paragraph-level annotations.
6. The system shall export annotations in a standards-compliant interoperable format.
7. The system shall store and query recursive literary-social relationships using Neo4j.
8. The system shall support discovery queries based on social graph relationships.

5.3 Non-Functional Requirements

- **Interoperability**

The component must align closely with ActivityPub and Web Annotation standards.

- **Performance**

The system should process core interactions and graph queries within practical real-time limits.

- **Scalability**

The architecture should support increasing volumes of users, relationships, and annotations.

- **Reliability**

Inbox/outbox communication and graph persistence should remain stable and consistent.

- **Privacy and Security**

Identity handling, content visibility, and signed activity delivery must be managed securely.

6.COMMERCIALIZATION PLAN

Target Market and Customer Persona

The target market includes:

- Online readers and book enthusiasts
- Academic institutions and students
- Digital bookstores and publishers
- Online reading communities

A typical customer persona could be a university student who regularly reads academic or literary books and wants to discuss ideas with other readers.

Value Proposition

The proposed system offers a **decentralized literary discussion platform** integrated into an online bookstore.

Unlike traditional bookstores where reviews remain confined to a single platform, this system allows readers to:

- Share reviews across decentralized networks
- Participate in open discussions
- Create collaborative annotations on book content

This enhances reader engagement and promotes knowledge sharing.

Pricing Strategy

The platform may adopt a **freemium model**.

Basic features such as browsing books and reading reviews will remain free for users.

Premium features may include:

- Advanced analytics for publishers
- Annotation collaboration tools for academic institutions
- Extended storage for annotations and discussions

Competitive Advantage

The proposed system provides several advantages over existing online bookstores:

- Decentralized social interaction using ActivityPub
- Graph-based analysis of reader interactions
- Collaborative annotation capabilities
- Cross-platform discussion integration

These features make the platform unique compared to traditional centralized book review systems.

Intellectual Property Considerations

The system architecture and integration approach may be protected through intellectual property mechanisms.

Potential protections include:

- Copyright for software implementation
- Licensing of the platform technology
- Protection of proprietary algorithms and system architecture

Open-source components used in the system will follow their respective licenses.

7. BUDGET AND JUSTIFICATION

The following table outlines the estimated budget required for the successful completion of the FedBook-Sem research component. The expenses have been categorized into hardware, software, cloud services, and human effort to ensure economic feasibility.

Description	Total Amount (Rs.)
Cloud Hosting for Web Application (Node.js backend and React frontend – 6 months)	10,000
Graph Database Hosting (Neo4j cloud service for social graph storage)	7,000
API Testing and Development Tools (Postman, development utilities)	2,000
Annotation Data Storage and Backup Services	3,000
ActivityPub Federation Testing Environment	3,500
Internet and Utilities for Development	3,000
User Interface Design Tools and Prototyping Software	2,500
Miscellaneous Expenses (unexpected costs, storage devices, adapters)	2,500
Total Budget	33,000

Table 4: Estimated Budget for the FedBook-Sem Component

7.1 Budget Justification Summary

The major cost of the FedBook-Sem component lies in deployment and graph data infrastructure rather than hardware. Because this component depends on federation testing and graph relationship processing, cloud hosting and Neo4j-related environments are more essential than physical devices. Development labour and workstation costs are minimized because the project will be completed using student-owned devices within the academic research scope.

8.WORK BREAKDOWN STRUCTURE (WBS)

Task ID	Task Description	Timeline	Deliverable
T1	Conduct literature review on decentralized social networks, ActivityPub protocol, and graph databases	March	Literature review report
T2	Analyse existing online bookstore review systems and identify limitations in centralized platforms	March April	Problem analysis report
T3	Design system architecture for the Federated Social Graph online bookstore platform	April	System architecture design
T4	Design graph database schema for users, books, reviews, comments, and annotations using Neo4j	May	Graph database schema
T5	Implement ActivityPub-based social interaction layer for federated communication	May – June	Federated interaction module
T6	Develop federated reviews, comments, likes, and shares functionality	June	Social interaction module
T7	Implement semantic annotation system based on W3C Web Annotation Data Model	July	Annotation management module
T8	Integrate Neo4j graph database to manage social relationships and interaction networks	August	Graph database implementation
T9	Develop web interface for review interaction and annotation features	September	Web application interface
T10	Perform system testing and evaluate performance of federated communication and graph queries	October	Testing and evaluation report
T11	Prepare final documentation and proposal presentation	November	Final research report

Table 5: FedBook-Sem | Work Breakdown Structure (WBS)

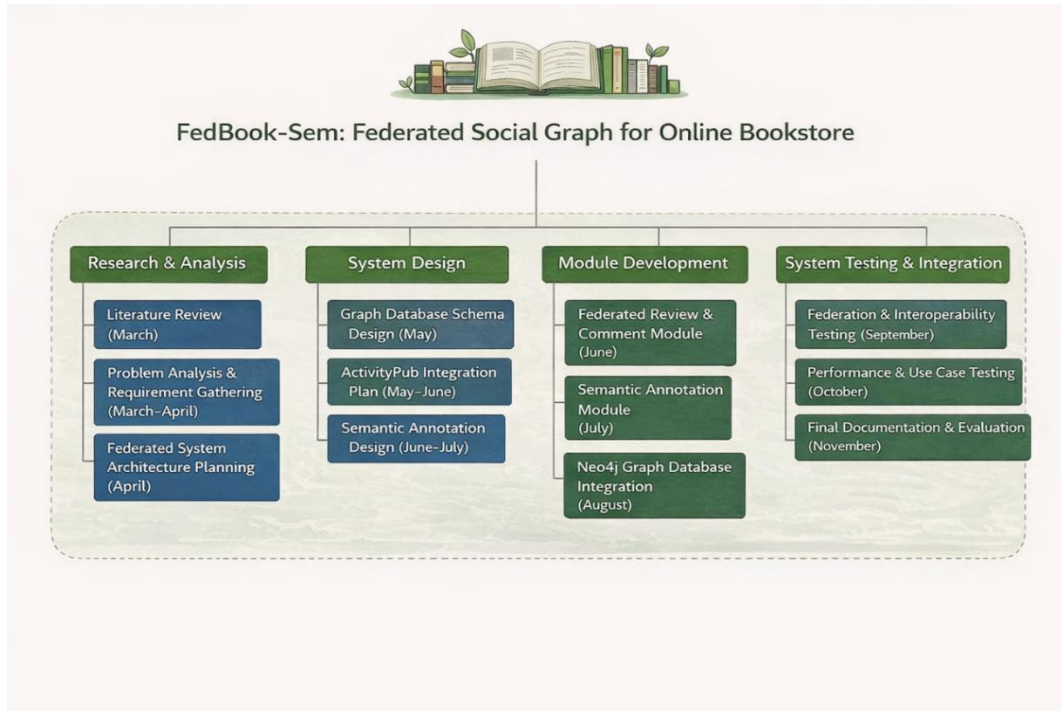


Figure 4: FedBook-Sem | Work Breakdown Structure (WBS)

9. TIME PLAN (GANTT CHART)

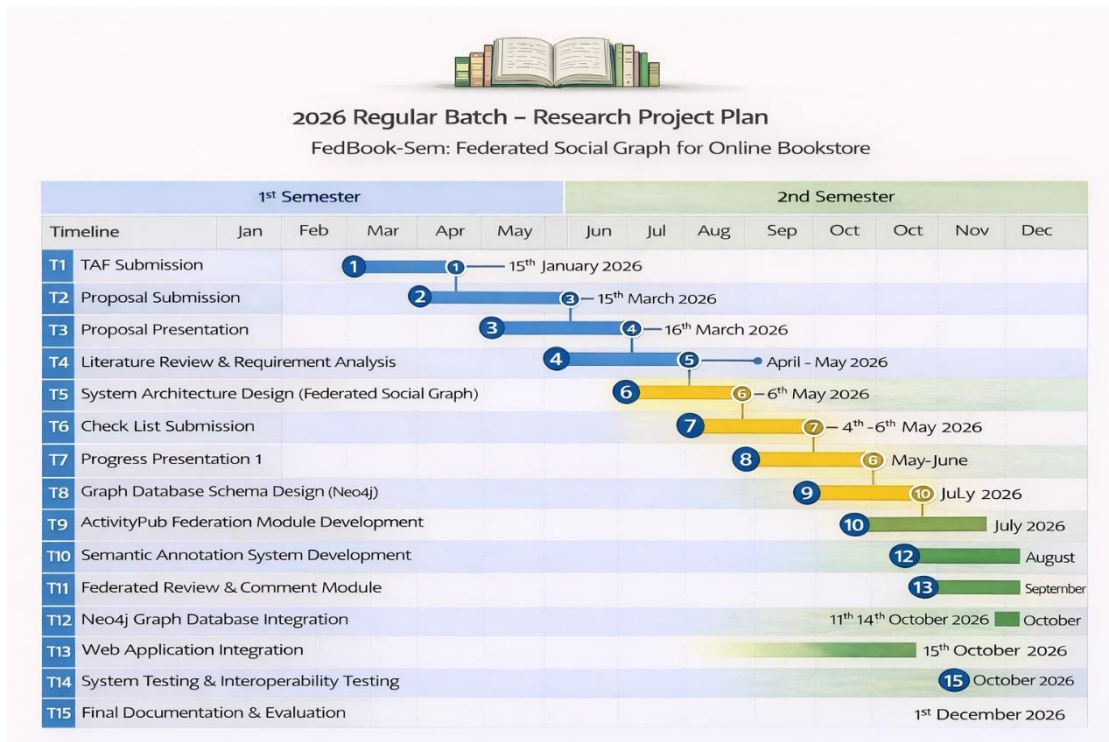


Figure 5: FedBook-Sem Research Project Gantt Char

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11. APPENDICES

11.1 Appendix A: Survey Instrument

Section A — Demographic Information

1. **What is your age group?**
 - Under 18
 - 18–24
 - 25–34
 - 35+
2. **How often do you read or purchase books?**
 - Very often
 - Occasionally
 - Rarely
 - Never

Section B — Reading Habits & Search Experience

3. **Which literary genres do you read or purchase most often?**
 - Fiction (Novels, Short Stories)
 - Academic/Educational books
 - Biographies/Autobiographies
 - Poetry
 - Self Help/Motivational books
 - Children’s Books
4. **How often do you face difficulty searching for Sinhala books online due to Unicode or spelling issues?**
 - Always
 - Sometimes
 - Rarely
 - Never

Section C — Search Preferences & Features

5. **Would you prefer using Voice Search or 'Singlish' typing (e.g., “Amba Yahaluwo” instead of “අම්බ යහලුවෝ”) to search for books?**
 - Yes – I highly prefer it
 - Maybe – I would like to try it
 - No – I am fine with current methods
6. **Have you ever struggled to find new books that match the specific “writing style” or emotional vibe of your favourite authors?**
 - Yes — frequently

- Sometimes
- Never

7. Rate how useful it would be to have a system that recommends books based on writing style and emotional tone rather than just popularity/sales:

(Scale 1–5: 1 Not useful — 5 Extremely useful)

Section D — Social Features & Reviews

8. How useful would it be to highlight specific sentences in an online book and share/discuss them with friends across platforms?

(Scale 1–5)

9. Would you like book reviews from social media to be integrated into the online bookstore experience?

- Yes – I like it
- Somewhat – It is slightly annoying
- No – I don't mind either way

Section E — Pricing and Purchasing Experience

10. Have you ever delayed buying an academic book or novel because of high prices during certain times (e.g., semester start)?

- Yes
- No

11. Would you be interested in a “Trade In Futures” feature that allows you to reserve books at a locked, affordable price before a semester begins?

- Yes, absolutely
- Maybe depending on price
- No

Section F — Discovery Sources

12. When looking for a new book, how much do you rely on the following sources for discovery?

Respondents rate each source: Not at all / Slightly / Moderately / Heavily

- Recommendations from friends/family
- Online bookstore recommendations (e.g., “Customers also bought”)
- Social media book reviews/influencers
- Physical browsing in bookstores

Section G — Open Ended Input

13. Please provide an example of a difficult Sinhala book title you had trouble finding online due to search issues.

Free text response field

Survey Use in Research

This survey was used to collect quantitative and qualitative data on user needs and preferences related to online book discovery. These insights helped justify the design of the TAF Online Bookstore system, particularly the need for:

- Singlish and Voice Search support
- Style based recommendation models
- Integration of social reviews
- Intelligent pricing features

Survey responses were stored securely and used only for research analysis. Participants were informed that the data would be confidential and used solely to develop system requirements.

11.2 Appendix B: Detailed Component Architecture Diagram

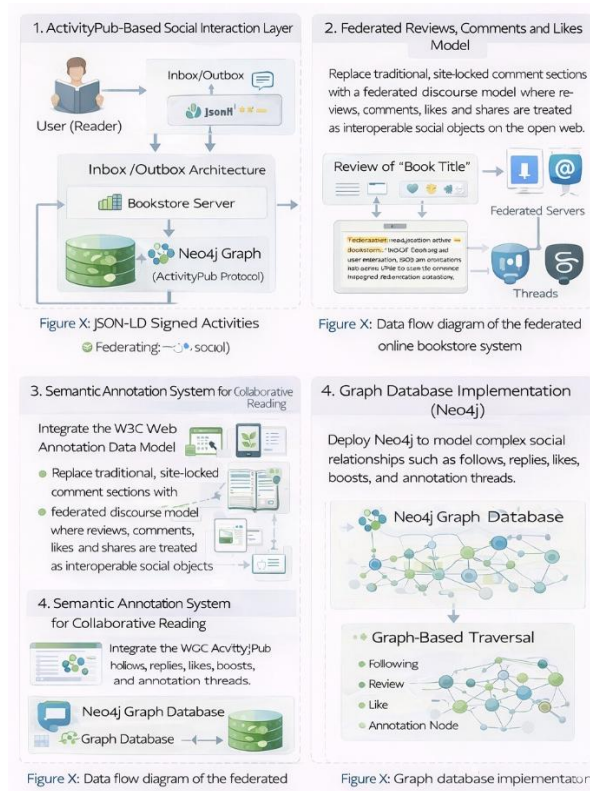


Figure 6: Detailed System Architecture of the FedBook-Sem Component

11.3 Appendix C: Preliminary UI mockups

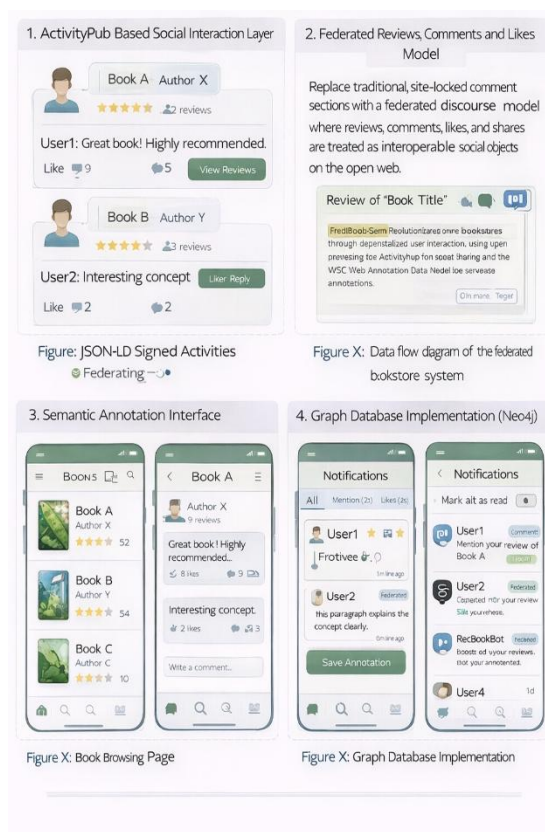


Figure 7: Preliminary UI mockups

11.4 Appendix D: Preliminary Risk Analysis and Mitigation Matrix

Risk ID	Risk Description	Impact	Mitigation Strategy
R1	Complexity in implementing the ActivityPub federation protocol	May cause interoperability issues between the bookstore system and external federated platforms	Use established ActivityPub libraries and follow W3C protocol standards; conduct interoperability testing with platforms such as Mastodon
R2	Synchronization issues between federated servers and the bookstore system	Inconsistent reviews, likes, or comments across platforms	Implement reliable message queues and activity synchronization mechanisms for federated interactions
R3	Performance limitations when querying large graph datasets in Neo4j	Slow response when retrieving relationships such as follows, likes, and replies	Optimize graph queries using indexing, caching mechanisms, and efficient traversal algorithms
R4	Security vulnerabilities in federated social interactions	Unauthorized activities or malicious data sharing across platforms	Implement authentication, access control mechanisms, and signed JSON-LD activities to ensure secure communication
R5	User adoption challenges for decentralized interaction features	Users may find federated systems complex compared to traditional platforms	Provide intuitive UI design and clear explanations of federated features within the bookstore interface
R6	Data privacy concerns related to user reviews, annotations, and interactions	Ethical and legal issues related to user-generated content	Implement user consent policies, data encryption, and privacy protection mechanisms for stored data

Table 6: Preliminary Risk Analysis and Mitigation Matrix

11.5 Appendix E: Ethical documents

The proposed research titled “**Federated Social Graph Architecture for Online Bookstore Platforms using ActivityPub and Graph Databases**” will follow ethical research standards throughout the system design, implementation, and evaluation phases. The research focuses on enabling decentralized social interaction for readers while ensuring privacy, security, and responsible use of user-generated data.

Informed Consent

Any user interactions such as **book reviews, comments, likes, shares, or annotations** collected for testing and evaluation will be obtained with the knowledge and consent of participants. Users involved in system testing will be informed about the purpose of the research and how their interaction data will be used for analysis. Participation will be voluntary, and users will have the option to withdraw from the study at any stage.

Data Privacy and Confidentiality

The system will not collect sensitive personal information beyond basic account data required for system functionality. Any user interaction data used for research evaluation will be **anonymized** to protect the identity of participants. Personal identifiers such as usernames or account details will not be disclosed in research reports or publications.

Responsible Use of User-Generated Data

User-generated content such as **reviews, comments, likes, and annotations** will be used strictly for academic research purposes and system evaluation. The research will respect content ownership principles of decentralized systems and will ensure that data shared through the **ActivityPub protocol** remains under the control of users and federated platforms.

Data Security

All system data including **user interactions, annotations, and social graph relationships** will be stored securely in the **Neo4j graph database**. Appropriate security measures such as authentication, access control, and encrypted communication protocols will be implemented to prevent unauthorized access or misuse of data.

Ethical Research Compliance

This research will comply with the **ethical research guidelines of the Sri Lanka Institute of Information Technology (SLIIT)**. The system will promote responsible digital communication and open knowledge exchange while ensuring that the implementation of decentralized social networking technologies does not harm individuals, communities, or organizations.